ZCAS University

CIT4251 – STRATEGIC IT MANAGEMENT

MID TERM TEST

23rd APRIL 2026

15.30 – 17.00 HOURS

TIME ALLOWED: ONE AND HALF HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Read the scenario in Appendix A and
2. Answer any **TWO** questions from three questions provided on the question paper. All questions carry 25 marks.
3. Candidates must **not turn this page** until the invigilator tells them to do so.

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Question 1.

a) Discuss the benefits and risks of adopting a hybrid cloud strategy. Explain how ZRG dealt with the risks of adopting a hybrid cloud strategy. (10 Marks)

b) Explain how virtualization contributes to efficient IT resource management. (5 Marks)

c) Assess the challenges organizations face when modernizing IT infrastructure. (10 Marks)

[Total: 25 Marks]

Question 2.

a) Critically evaluate the impact of enterprise social media on organizational communication. (10 Marks)

b) Identify and explain the risks associated with enterprise social media. (10 Marks)

c) Analyse strategies to improve user adoption of collaboration technologies. (5 Marks)

[Total: 25 Marks]

Question 3.

a) Differentiate between data warehouses and big data platforms. (5 Marks)

b) Evaluate challenges in implementing big data analytics. (10 Marks)

c) Assess how business intelligence systems create competitive advantage. (10 Marks)

[Total: 25 Marks]

APPENDIX A:

Case Study: Digital Transformation at Zambezi Retail Group (ZRG)

Case Background

Zambezi Retail Group (ZRG) is a fast-growing retail chain operating across Southern Africa, with headquarters in Lusaka. The company specializes in consumer goods, groceries, and household electronics. Over the past decade, ZRG has expanded rapidly, but its Information Systems (IS) infrastructure has not evolved at the same pace.

ZRG's legacy systems consist of fragmented databases, standalone inventory applications, and minimal integration between branches. Communication across departments relies heavily on email and manual reporting, leading to delays and inefficiencies. Furthermore, the company lacks advanced analytics capabilities, limiting its ability to leverage customer data for strategic decision-making.

Facing increasing competition from digitally mature retailers and e-commerce platforms, ZRG's executive board approved a strategic IT transformation initiative. The initiative focuses on three core areas, **Modernizing IT Infrastructure and Services, Enhancing Organizational Communication via Social Media Tools and Leveraging Big Data and Analytics for Business Intelligence.**

ZRG's IT department identified several infrastructure challenges, four of which were prominent, aging on-premise servers with frequent downtime, lack of scalability to support growing operations, weak disaster recovery mechanisms and high maintenance costs.

To address these issues, ZRG adopted a **hybrid cloud strategy**, migrating critical applications such as inventory management and financial systems to cloud platforms while retaining sensitive data on-premises.

Additionally, ZRG implemented **IT Service Management (ITSM)** frameworks aligned with ITIL principles, Centralized data centers for improved system reliability and Virtualization technologies to optimize resource utilization. This transformation improved system availability, reduced operational costs, and enabled faster deployment of new services.

ZRG recognized that poor communication was affecting productivity and decision-making. To address this, the company deployed enterprise social media and collaboration tools, including internal communication platforms (e.g., Slack-like systems), corporate social networks for knowledge sharing and video conferencing tools for remote collaboration. These tools enabled real-time communication across geographically dispersed branches, enhanced knowledge sharing through discussion forums and improved employee engagement and innovation

Despite these actions, challenges emerged, including information overload, security concerns related to data sharing and resistance to change among employees

ZRG responded by implementing governance policies, training programs, and usage guidelines to ensure effective adoption.

ZRG generates vast amounts of data from point-of-sale (POS) systems, customer loyalty programs, supply chain operations and social media interactions. Previously, this data was underutilized. As part of its transformation, ZRG implemented a **data warehouse** for structured data integration, **Big Data platforms** (e.g., Hadoop ecosystem concepts) for handling unstructured data and a**dvanced analytics tools** for predictive and prescriptive insights.

The key outcomes from these implementations included improved demand forecasting, personalized marketing campaigns, optimized inventory management and enhanced customer experience. For example, analytics revealed purchasing patterns that allowed ZRG to tailor promotions based on regional preferences, significantly increasing sales.

Conclusion

ZRG's digital transformation demonstrates the strategic importance of integrating IT infrastructure, communication systems, and analytics capabilities. By aligning IT with business goals, ZRG achieved operational efficiency, improved collaboration, and data-driven decision-making—key components of sustainable competitive advantage.